

Rule 37 under fixed sequential updates: a minimum six-cell cover and a resistant twelve-cell state

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

We establish two exact finite results for elementary cellular automaton Rule 37 with an in-place sequential order repeated at every sweep. On six cells, the minimum number of fixed orders whose fixed-point basins cover all configurations is three; an explicit attaining family reaches a fixed point from every state within two sweeps. One of these orders also gives a two-sweep counterexample to the six-cell resistance assertion in Section 3.3.1 of arXiv:2509.07797, versions 1 and 2. On twelve cells, the state 100100010000 avoids all fixed points for every fixed order and every nonnegative sweep time. Commutation of nonadjacent updates reduces order coverage to 62 and 4094 acyclic cycle orientations, respectively. Complete orbit certificates and standalone exact verification code accompany the proofs. The conclusions concern these two finite sizes; no general-size or literature-priority claim is made.

1 Model, context, and results

For $n \in \{6, 12\}$, let $V_n = \mathbb{Z}/n\mathbb{Z}$ and $X_n = \{0, 1\}^n$. Coordinates are labelled $0, \dots, n-1$, and every displayed word has coordinate zero first. The local rule $f : \{0, 1\}^3 \rightarrow \{0, 1\}$ is

(a, b, c)	111	110	101	100	011	010	001	000
$f(a, b, c)$	0	0	1	0	0	1	0	1.

The triple consists of the left neighbour, centre, and right neighbour. An update U_i replaces only coordinate i by $f(x_{i-1}, x_i, x_{i+1})$, with indices modulo n and all inputs read from the current state. For a permutation $\pi = (\pi_1, \dots, \pi_n)$, define

$$F_\pi = U_{\pi_n} \circ \dots \circ U_{\pi_1}.$$

The iterate F_π^t repeats the same order at every sweep. Write

$$\mathcal{P}_n = \{(001)^{n/3}, (010)^{n/3}, (100)^{n/3}\}, \quad B_\pi = \{x \in X_n : \exists t \in \mathbb{Z}_{\geq 0}, F_\pi^t(x) \in \mathcal{P}_n\}.$$

As shown below, $\mathcal{P}_n$ is exactly the fixed-point set of every sweep. A family $\mathcal{S} \subseteq S_n$ is a *cover* if $\bigcup_{\pi \in \mathcal{S}} B_\pi = X_n$. Each initial state chooses one member of the family and repeats that member forever; there is no switching of orders along a trajectory.

Theorem 1. *On six cells, the minimum cardinality of a cover is exactly three. An attaining family consists of*

$$A = (0, 1, 2, 5, 4, 3), \quad B = (0, 1, 4, 3, 2, 5), \quad C = (0, 3, 2, 1, 4, 5).$$

For every $x \in X_6$, there are $\pi \in \{A, B, C\}$ and $t \in \{0, 1, 2\}$ such that $F_\pi^t(x) \in \mathcal{P}_6$. For every pair $\pi, \sigma \in S_6$, including $\pi = \sigma$, there is a state outside both B_π and B_σ .

Theorem 2. On twelve cells, let $x = 100100010000$, with support $\{0, 3, 7\}$. Then

$$\forall \pi \in S_{12} \quad \forall t \in \mathbb{Z}_{\geq 0}, \quad F_\pi^t(x) \notin \mathcal{P}_{12}.$$

Consequently no family of fixed sequential orders covers X_{12} .

The motivating reference is Donoso-Leiva, Góes, Ríos-Wilson, and Sené [1]. Their finite-ring model and covering definition agree with the convention above: see version 2, Sections 2.1–2.2 and Definition 4. Their Rule 37 discussion in Section 3.3.1 reports a six-cell configuration that resists every sequential mode. Section 4 below corrects that specific statement by a directly checkable trajectory. Theorem 1 gives the stronger finite conclusion that three modes are necessary and sufficient to cover all six-cell states, answering this instance of the minimal-cover question in their version 2, Section 4. Theorem 2 resolves the specified twelve-cell existence question and provides a contrasting size at which a cover fails.

Neither of the two inspected Rule 37 sections states these two finite results. This comparison is limited to arXiv versions 1 and 2; it does not exclude later corrections or independent prior occurrences. The printed Conjecture 2 does not explicitly specify its ring-size quantifier. We do not choose a uniquely intended global reading, and make no claim about its general or eventual-size formulations, or the reported nine-cell example. The orientation reduction used here is standard update-order machinery and is proved directly, without a claim of novelty. For adjacent cellular-automaton period context, see Zeng [2], which concerns reversible second-order Rule 115 and supplies no premise for the Rule 37 proofs below.

2 Fixed points and complete order coverage

Lemma 3. For $n \in \{6, 12\}$ and every $\pi \in S_n$, the fixed-point set of F_π is exactly $\mathcal{P}_n$. Each of these states is fixed by every U_i .

Proof. In a locally fixed state, a central one must have neighbourhood 010, and a central zero must have neighbourhood 001 or 100. The all-zero state is not locally fixed, since $f(0, 0, 0) = 1$. Thus there is a one, all ones are isolated, and every zero has exactly one adjacent one. Between consecutive ones there cannot be one zero, which would have two adjacent ones, or three or more zeros, which would include a zero with no adjacent one. There are exactly two zeros, giving precisely $\mathcal{P}_n$. Conversely, its three states satisfy each local fixed-point condition.

If $F_\pi(y) = y$, each coordinate is written exactly once during the sweep, so its sole write must give its original value. The first update therefore leaves y unchanged. Inductively so does every subsequent update, proving that y is locally fixed. The reverse implication is immediate. $\square$

In particular, reaching a target during a sweep also reaches it at the next sweep boundary, because all subsequent updates preserve that target.

If i and j are distinct nonadjacent cells, then $U_i U_j = U_j U_i$. Indeed, neither writes any input read by the other. Their read sets may overlap, but their written coordinates lie outside each other's read sets. Orient each cycle edge from its earlier to its later endpoint in an order. Encode the orientation by

$$m(\pi) = \sum_{i=0}^{n-1} 2^i \mathbf{1}\{i \text{ occurs before } i+1 \pmod{n} \text{ in } \pi\}. \quad (1)$$

The last comparison is between cells $n-1$ and 0.

Lemma 4. *The possible orientation masks are precisely $1, \dots, 2^n - 2$. Each has a representative permutation. All orders with the same mask have the same sweep function on every state of X_n .*

Proof. Ranks in a permutation increase along every directed edge, so its orientation is acyclic. Conversely, an acyclic orientation has a topological order: repeatedly remove a vertex of indegree zero. Such a vertex exists, since otherwise following predecessors would eventually give a directed cycle. A cycle graph has no proper undirected cycle, so a directed cycle in an orientation must use every edge. Only the two consistently directed orientations are cyclic, with masks 0 and $2^n - 1$.

Orders inducing one orientation are the linear extensions of its reachability partial order. Any two extensions are connected by swaps of consecutive incomparable vertices. To see this, move the first vertex of the desired extension leftwards in the current one. Every crossed vertex is incomparable: a smaller vertex would precede it in the desired extension, and a larger vertex could not precede it in the current one. Repeat on the suffix. Vertices joined by a cycle edge are comparable, so every swap exchanges nonadjacent cells and hence commuting updates. Each swap preserves the sweep function. $\square$

This covers all permutations using 62 masks for $n = 6$ and 4094 for $n = 12$. Distinct masks need not induce distinct functions. No rotation or reflection of a state, or cyclic shift of an update sequence, is identified.

3 Exact finite certificates

A state is encoded by

$$\text{enc}(y) = \sum_{i=0}^{n-1} y_i 2^i. \quad (2)$$

Thus the targets are $\{9, 18, 36\}$ for $n = 6$ and $\{585, 1170, 2340\}$ for $n = 12$. The twelve-cell witness is 137. Displayed words have the coefficient of 2^0 on the left, opposite to ordinary binary integer notation.

Lemma 5. *Let F be a deterministic sweep. Suppose distinct states $s_0, \dots, s_{L-1}$ avoid $\mathcal{P}_n$, and $0 \leq \mu < L$, with*

$$F(s_j) = s_{j+1} \quad (0 \leq j < L - 1), \quad F(s_{L-1}) = s_\mu. \quad (3)$$

Then $F^t(s_0) \notin \mathcal{P}_n$ for every integer $t \geq 0$.

Proof. The finite listed set contains s_0 and is forward invariant, including its last state. Induction keeps every iterate in that set. The first μ states form a transient and the remaining $L - \mu$ form a cycle. $\square$

We call such a record a *closed orbit certificate*, or lasso. Every closing edge is checked explicitly; no simulation cutoff certifies infinite-time avoidance.

3.1 Six cells: the cover and its minimum size

The supplement's `n6-cover-certificate.json` contains three parts:

1. For each mask $1 \leq m \leq 62$, a representative order, its full 64-state sweep map, and its basin of the three targets.
2. For each of the 64 initial states, a trace under one of the three cover masks 7, 19, 25, ending in a target within two sweeps.

3. For each pair $1 \leq a \leq b \leq 62$, a state outside both basins.

The file `n6-pair-escape-orbits.jsonl` provides the closed orbits needed for the last part. There are $62 \cdot 63/2 = 1953$ pairs with repetition. After sharing identical requests for the same mask and initial state, 335 closed orbits suffice. Each row specifies `mask`, `initial_state`, `states`, and `repeat_index`; the last two fields encode the list and μ in Lemma 5. The verifier checks the exact pair domain, the required orbit domain, state bounds, initial states, target avoidance, all sweep transitions, and every closing index and edge.

For a separate exhaustive check of the order reduction at this size, the verifier enumerates all $6! = 720$ actual permutations and evaluates their sweep maps on all 64 states. These 46,080 full sweeps agree with the maps stored for their respective masks. Basins can then be reconstructed without a time bound: start with the targets and repeatedly add all predecessors under the sweep map until the set stops growing. This set is exactly the basin, since membership is equivalent to reaching a target along a finite directed path.

The recorded checks give basin sizes 40, 39, 40 for A, B, C , respectively, whose masks are 7, 19, 25. Their union is all 64 states. The 64 upper-bound traces use 96 transitions in total; all have at most two transitions. The 335 escape lassos use 2744 transitions including closures. Every one of the 1953 class pairs has the required escaping state. As a redundant check, the largest union of two reconstructed basins has 59 states.

Proof of Theorem 1. The 64 verified upper traces prove that $\{A, B, C\}$ is a cover with the stated two-sweep bound, using Lemma 4 to identify any stored representative with its listed attaining order. For every class pair, the two verified closed orbits from its assigned state and Lemma 5 prove avoidance under both representatives for all time. Lemma 4 transfers this avoidance to any two orders in those classes. Thus no pair covers X_6 . Repeated class pairs also exclude singleton covers, and the empty family does not cover the nonempty state space. The minimum is therefore exactly three. $\square$

3.2 Twelve cells: resistance to every order

The file `n12-witness-orbits.jsonl` contains exactly one row for each mask $1, \dots, 4094$. Each row specifies `mask`, `order`, `states`, and `repeat_index`. The verifier checks that the order is a permutation with the stated mask; the state list is nonempty, consists of distinct integers in $[0, 4095]$, and starts at 137; no listed state is a target; and the closing index and every transition satisfy (3). All 4094 rows pass, with 64,193 transitions including closures. The largest list has 26 states. Table 1 gives the period counts. They count orientation representatives, not permutations or resistant states.

Table 1: Periods of the certified twelve-cell witness orbits.								
Period	3	13	16	17	19	20	22	23
Representatives	1364	13	794	434	1150	196	132	11

For example, mask 1 has representative $(0, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1)$ and state list

137, 1207, 1242, 1132, 2615, 3354, 1677, 2887, 3490,
1745, 2921, 3508, 1754, 876, 439, 2266, 1133.

The last state maps to 2615, at index 4, closing a period-13 cycle. This example illustrates the format; the verification uses the complete file.

Proof of Theorem 2. Every representative has a verified closed orbit from 137 avoiding the targets. Lemma 5 proves avoidance for every nonnegative time. Every permutation has one of the checked masks and the same sweep map as its representative by Lemma 4. Their iterates therefore agree. This proves both universal quantifiers. Since the witness is in none of the basins, no family of orders can cover the state space. $\square$

4 The specific primary-source correction

In arXiv:2509.07797v2, Section 3.3.1, page 35, the authors state that for $n = 6$, the configuration 001000 “was unable to reach a fixed point regardless of the sequential update mode” [1]. The same assertion occurs in version 1, Section 3.3.1, printed page 33 (PDF page 34). The rule table is Table 16 in version 2 and Table 17 in version 1. Both tables and the stated sequential conventions agree with those used here.

Under the fixed order $A = (0, 1, 2, 5, 4, 3)$, direct evaluation gives

$$001000 \xrightarrow{F_A} 110010 \xrightarrow{F_A} 010010.$$

For complete local verification, the first sweep has the following states immediately after its six updates, including unchanged substeps:

updated cell	0	1	2	5	4	3
state	101000	111000	110000	110000	110010	110010.

The first update of the second sweep changes 110010 to 010010; its remaining five updates leave this state unchanged. The final state belongs to $\mathcal{P}_6$, so the printed six-cell resistance assertion is false under the stated model. This direct counterexample does not depend on exhaustive computation. It does not identify the cause of the reported simulation error or address the nine-cell assertion.

5 Reproduction and limits

The supplement includes all three certificates and a standalone Python 3 program `replay.py`, using only the standard library. Run

```
python3 manuscript/supplement/replay.py
```

from the project root, or invoke the program by its absolute path from any directory. It reads its adjacent certificates, writes no files, and prints a JSON report. Any failed condition raises an exception.

The packaged verifier uses tuple-valued states and the literal eight-entry rule table. It reconstructs six-cell basins by reverse reachability, checks all 720 permutations and all 259,560 unordered permutation pairs with repetition, and replays the upper and lower certificates. For twelve cells, it also constructs a largest-available-first topological order directly from each mask and follows the witness to its first exact repeat. The regenerated lists and repeat indices match the certificate, checking another 64,193 transitions. These alternate orders differ from the recorded ones in 4070 cases.

This implementation was written in a separate mathematical review pass without importing the certificate generator or the research verifier. For publication, only paths, output handling, and internal provenance handling were changed. The original generators used integer bit operations;

a separate research verifier used strings and the literal local table. The included replay is a reproducibility artifact, and rerunning it does not constitute an additional independent review.

The results use exact finite Boolean arithmetic, with no floating-point inference. They rely on the stated mathematical reductions and complete finite certificate checks, with the usual dependence on correct execution of the verifier and runtime. They are not proof-assistant formalizations. The twelve-cell existential question is resolved, but no complete classification of resistant states, theorem for other sizes, or result for orders varying between sweeps is asserted. Later errata and independent prior occurrences have not been excluded.

Computational and AI assistance. OpenAI Codex assisted with finite search, certificate generation, verification code, primary-source comparison, proof exposition, and manuscript preparation. A separate model context reconstructed the combined argument and wrote the tuple-rule verifier described above. This was a separate computational review pass, not external peer review or formal proof-assistant certification.

References

- [1] Isabel Donoso-Leiva, Eric Goles, Martín Ríos-Wilson, and Sylvain Sené, *On the Convergence of Elementary Cellular Automata under Sequential Update Modes*, arXiv:2509.07797, version 1 (9 September 2025), version 2 (24 February 2026). <https://arxiv.org/abs/2509.07797v1>; <https://arxiv.org/abs/2509.07797v2>.
- [2] Zijian Zeng, *Exact Odd-Parameter Periods for Reversible Elementary Second-Order Cellular Automaton Rule 115*, 2026. <https://doi.org/10.2139/ssrn.7379379>.